

Estimation of sample size

Estimation of Sample Size

1. Introduction

In research, it is usually **not possible or practical to collect data from every member of a population**. Therefore, researchers select a smaller group called a **sample** and use the information obtained from this sample to draw conclusions about the entire population.

However, an important question arises:

How many observations or respondents should be included in the sample?

The process of determining the appropriate number of observations required for a study is called **estimation of sample size**.

An appropriately estimated sample size helps the researcher obtain **reliable and statistically meaningful results** while avoiding unnecessary time, cost, and effort.

2. Definition

Sample size estimation is the statistical process of determining the **minimum number of observations, participants, or units that should be included in a sample** to obtain results that are sufficiently reliable and representative of the population.

Simple meaning

Sample size = Number of individuals or observations required in a research study.

For example, if a university has **10,000 students** and a researcher wants to study students' satisfaction with online learning, it may not be necessary to survey all 10,000 students. The researcher may calculate that a sample of **370 students** is sufficient.

3. Why is Sample Size Important?

Choosing the correct sample size is important because it directly affects the quality of research findings.

1. Reliability of results

A sufficiently large sample generally provides more stable estimates of population characteristics.

2. Accuracy

A properly calculated sample reduces the possibility that the findings are substantially affected by random sampling variation.

3. Cost and time

Studying the entire population can be expensive and time-consuming. An appropriate sample reduces these requirements.

4. Statistical significance

Sample size affects the ability of statistical tests to detect meaningful differences or relationships.

5. Generalization

A properly selected sample allows researchers to make reasonable inferences about the population from which it was drawn.

4. Factors Affecting Sample Size

The required sample size does not depend on population size alone. Several factors must be considered.

A. Population Size

Population size refers to the total number of units in the population.

For example:

- Students in a university = 10,000
- Employees in an organization = 2,000
- Customers of a company = 50,000

For very large populations, the required sample size often changes relatively little once the population becomes sufficiently large.

B. Level of Confidence

The **confidence level** indicates how confident the researcher wants to be that the estimated result represents the population.

Common confidence levels are:

Confidence Level	Z-value
90%	1.645
95%	1.96
99%	2.576

A higher confidence level generally requires a larger sample.

For example:

A study requiring 99% confidence will generally require more observations than one requiring 90% confidence, assuming the other conditions remain the same.

C. Margin of Error

The **margin of error** indicates how much the sample estimate may differ from the actual population value.

Common choices include:

- $\pm 10\%$
- $\pm 5\%$
- $\pm 3\%$

A smaller margin of error requires a larger sample.

For example:

- $\pm 10\% \rightarrow$ smaller sample
- $\pm 5\% \rightarrow$ larger sample
- $\pm 3\% \rightarrow$ still larger sample

Thus:

Higher precision $\rightarrow$ Larger sample size

D. Population Variability

Population variability refers to how much the characteristics of individuals differ from one another.

If variability is high, a larger sample may be required to obtain a reliable estimate.

When the population proportion is unknown, researchers often use:

$$p = 0.5$$

because it gives the maximum variability and therefore provides a conservative sample-size estimate.

E. Nature of the Study

Different types of research may require different sample sizes.

For example:

- Descriptive survey
- Experimental research
- Correlation study
- Regression analysis
- Clinical research

The statistical method and research design influence the required sample size.

F. Expected Effect Size

In hypothesis-testing studies, **effect size** represents the magnitude of the difference or relationship that the researcher wants to detect.

A small expected effect generally requires a larger sample.

A large expected effect may be detectable with a smaller sample.

G. Statistical Power

Statistical power is the probability that a statistical test will detect an effect when a real effect exists.

A commonly used value is:

$$Power = 80\%$$

Higher statistical power generally requires a larger sample.

5. Methods of Estimating Sample Size

There are several methods for estimating sample size.

Common methods include:

1. Slovin's formula
2. Yamane's formula
3. Cochran's formula
4. Krejcie and Morgan table
5. Power analysis
6. Statistical software such as **G*Power**

The appropriate method depends on the research design and statistical analysis.

6. Slovin's Formula

One commonly taught formula for estimating sample size when the population size is known is:

$$n = \frac{N}{1 + Ne^2}$$

Where:

- n = required sample size
 - N = population size
 - e = acceptable margin of error
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Example

Suppose a college has:

$$N = 10,000$$

students and the researcher wants a margin of error of:

$$e = 5\% = 0.05$$

Using Slovin's formula:

$$n = \frac{10,000}{1 + (10,000)(0.05)^2}$$

First calculate:

$$(0.05)^2 = 0.0025$$

Therefore:

$$n = \frac{10,000}{1 + (10,000)(0.0025)}$$

$$n = \frac{10,000}{1 + 25}$$

$$n = \frac{10,000}{26}$$

$$n = 384.62$$

Since the sample size must be a whole number:

$$n \approx 385$$

Answer:

The researcher should select approximately **385 students**.

7. Cochran's Formula

Cochran's formula is commonly used when estimating a population proportion, particularly for large populations.

$$n_0 = \frac{Z^2 pq}{e^2}$$

Where:

- n_0 = initial sample size
- Z = Z-value corresponding to confidence level
- p = estimated proportion of the population
- $q = 1 - p$
- e = margin of error

If the population proportion is unknown, researchers commonly use:

$$p = 0.5$$

Therefore:

$$q = 1 - 0.5 = 0.5$$

Numerical Example

Suppose a researcher wants:

- Confidence level = 95%
- Margin of error = 5%
- $p = 0.5$

For 95% confidence:

$$Z = 1.96$$

Therefore:

$$n_0 = \frac{(1.96)^2(0.5)(0.5)}{(0.05)^2}$$

$$n_0 = \frac{3.8416 \times 0.25}{0.0025}$$

$$n_0 = \frac{0.9604}{0.0025}$$

$$n_0 = 384.16$$

Therefore:

$$\boxed{n \approx 385}$$

Thus, approximately **385 respondents** are required.

8. Finite Population Correction

When the population is relatively small and its exact size is known, the initial Cochran sample size can be adjusted.

The formula is:

$$n = \frac{n_0}{1 + \frac{n_0 - 1}{N}}$$

Where:

- n = adjusted sample size
- n_0 = initial sample size

- N = population size

Example

Suppose:

$$n_0 = 385$$

and:

$$N = 1,000$$

Then:

$$n = \frac{385}{1 + \frac{384}{1000}}$$

$$n = \frac{385}{1.384}$$

$$n \approx 278$$

Therefore:

$$\boxed{n \approx 278}$$

For a population of 1,000, approximately **278 respondents** would be required under these assumptions.

9. Sample Size and Non-Response

Researchers should also consider that some selected participants may:

- refuse to participate,
- submit incomplete questionnaires,
- drop out of the study, or
- fail to respond.

Therefore, researchers may increase the number of participants initially approached.

Example

Suppose the required sample is:

$$n = 385$$

and the researcher expects a **10% non-response rate**.

A simple adjustment is:

$$\begin{aligned} n_{adjusted} &= \frac{385}{1 - 0.10} \\ &= \frac{385}{0.90} \\ &= 427.78 \end{aligned}$$

Therefore:

428

The researcher may approach approximately **428 participants** to obtain about 385 usable responses.

10. Sample Size in Different Situations

Situation	Important consideration
Large population survey	Confidence level, margin of error, variability
Small population	Finite population correction
Experimental study	Effect size and statistical power
Clinical study	Effect size, power, significance level, dropout
Regression study	Number of predictors and desired power
Qualitative research	Depth of information and saturation
Questionnaire survey	Expected response rate

11. Sample Size vs Sampling Technique

It is important to distinguish between **sample size** and **sampling technique**.

Sample size

Answers:

How many people should be selected?

Example:

$$n = 385$$

Sampling technique

Answers:

How should those people be selected?

Examples:

- Simple random sampling
- Stratified sampling
- Systematic sampling
- Cluster sampling
- Quota sampling
- Snowball sampling

Thus, a researcher may decide that **385 students** are required and then use **stratified random sampling** to select those students.

12. Practical Scenario

Scenario: Student Satisfaction Survey

A university has **10,000 students**. The administration wants to study student satisfaction with online learning.

The researcher decides:

- Population = 10,000
- Confidence level = 95%
- Margin of error = 5%

Using Cochran's formula:

$$n_0 = \frac{Z^2 pq}{e^2}$$
$$n_0 = \frac{(1.96)^2 (0.5)(0.5)}{(0.05)^2}$$
$$n_0 \approx 385$$

Therefore, the researcher needs approximately **385 valid responses**.

If a 10% non-response rate is expected:

$$n_{adjusted} = \frac{385}{0.90}$$
$$\approx 428$$

Therefore, the researcher may invite approximately **428 students** so that around **385 usable responses** are obtained.

13. Important Relationship

A useful way to remember the concept is:

Higher confidence $\Rightarrow$ Larger sample

Smaller margin of error $\Rightarrow$ Larger sample

Higher variability $\Rightarrow$ Larger sample

Higher statistical power $\Rightarrow$ Larger sample

Higher expected response rate $\Rightarrow$ Smaller number initially contacted

14. Advantages of Proper Sample Size Estimation

1. Improves **reliability** of research findings.
 2. Reduces **sampling error**.
 3. Saves **time and money** compared with studying the entire population.
 4. Helps researchers achieve **adequate statistical power**.
 5. Prevents unnecessarily large samples.
 6. Supports better **generalization** of findings.
 7. Makes the research design more systematic and scientifically justified.
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15. Conclusion

Estimation of sample size is an important step in research methodology. It determines how many observations or participants are needed to obtain reliable results from a population.

The required sample size depends on factors such as **population size, confidence level, margin of error, population variability, effect size, statistical power, research design, and expected non-response.**

For example, using Cochran's formula with a **95% confidence level, 5% margin of error, and $p = 0.5$** gives an initial sample size of approximately **385** for a large population.

Therefore, proper sample-size estimation helps the researcher achieve a balance between **statistical reliability and practical feasibility**.



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